Geometric boundaries of the Frohmanian symplectic tether: formal tracking of the Navier–Stokes millennium problem

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October 2026
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Abstract

This note records the boundary of the Frohmanian symplectic tether as applied to Fefferman's Statement (B), unforced existence and smoothness on $\mathbb{T}^3$. The tether is a bilinear antisymmetric correction $B(F, G)$ built so that an L^2-orthogonal projection satisfies $\Pi_u(u) = 0$, and hence $B(F, H) \equiv 0$ on the kinetic energy. That identity is schematic in the accompanying Lean tree: it is a field of a structure, not a derivation of the bilinear form on $\mathbb{T}^3$. It does not supply an a priori bound on $\|\omega\|_{L^\infty}$. A model of a steady two-dimensional Beltrami field has constant direction, so the Constantin–Fefferman bound holds, while the enstrophy clock grows linearly. A linearly growing history is not a divergent Beale–Kato–Majda integral. Statement (B) remains open.

1 The geometric core

The intended setting is the coadjoint orbit of volume-preserving diffeomorphisms of $\mathbb{T}^3$, with the unmodified incompressible equations. The tether is a bilinear antisymmetric correction $B(F, G)$ using an L^2-orthogonal projection Π_u onto the complement of the velocity. The defining identity is

$$\Pi_u(u) = 0.$$

On the kinetic energy H, with the formal identification $\delta H/\delta\omega = u$, the correction is built to vanish:

$$B(F, H) \equiv 0.$$

In the repository this is `tether_degeneracy` and `tether_bracket_preservation`, both fields of a structure. They do not bound $M(t) = \|\omega(t)\|_{L^\infty}$ and they do not estimate

$$\alpha = \xi \cdot \nabla u \cdot \xi.$$

A form that contributes nothing on H does not absorb topology-preserving stretching. That precludes this bracket from being the missing a priori bound. It does not preclude a separate analytic estimate from bounding α.

2 The enstrophy clock

The auxiliary field used in the Section 10 bootstrap satisfies

$$\partial_t \phi + u \cdot \nabla \phi = |\omega|^2, \qquad \phi(0, x) = 0,$$

and therefore $\|\phi(t)\|_{L^\infty} \le \int_0^t M(s)^2\,ds$. For a steady Beltrami field, $\nu = 0$ and $M(t) \equiv M_0 > 0$,

$$\|\phi(t)\|_{L^\infty} = M_0^2 t.$$

A z-independent field has $\omega = \omega_3 e_3$, so ξ is constant and $\sin\varphi = 0$, which is the Constantin–Fefferman bound for every $\rho > 0$. Global regularity of viscous two-dimensional Navier–Stokes is classical (Ladyzhenskaya) and is cited, not re-proved. The Lean file `TwoDClock.lean` defines $\sin\varphi$ to be 0 and the clock to be $M_0^2 t$, closed by reflexivity. That models the collision. It does not construct the Beltrami field.

The collision disqualifies a majorant whose positive term contains $\int_0^t M^2$. On a solution that is already regular, that coefficient grows like $t^{3/2}$ in the pasted driver. The autonomous comparison $y' = Cy^2 - \kappa'' y^3$ is not a majorant. Viscous two-dimensional solutions decay, so this is not a claim that $\int_0^\infty M^2$ diverges.

3 Alignment does not delete the clock

Constantin–Fefferman assumes $|\sin\varphi(x, x+y, t)| \le |y|/\rho$ where both vorticities are large, and concludes regularity. The hypothesis controls α through the Biot–Savart kernel. The clock integrates $|\omega|^2$ and does not read ξ. The same model has both: $\sin\varphi = 0$ and $\|\phi(t)\|_{L^\infty} = M_0^2 t$. Substituting the angle bound into the old Lyapunov functional does not remove the history.

Deriving the angle bound from $B(F, H) \equiv 0$ is a type error. The bracket quantifies over functionals. The angle bound quantifies over $\xi(x) - \xi(y)$. Proving the angle bound for every smooth divergence-free initial field is Statement (B) again. The alignment module stays conditional. No scale-critical weight $Z = |\omega|^{-1/4}\omega$ is extracted from the tether.

4 The Beale–Kato–Majda boundary

The criterion requires

$$\int_0^T M(t)\,dt < \infty.$$

In the steady model this integral equals $M_0 T$, finite on every compact interval, in agreement with regularity. Linear growth of $\int_0^T M^2\,ds = M_0^2 T$ is a different integral, also finite on that interval. It is not a divergent Beale–Kato–Majda integral, and it is not a blowup proof. The obligation $\int_0^{T_*} M(t)\,dt < \infty$ with a bound depending only on the data stays an open proposition in the Lean tree. It is not defined to be false.

Statement (B) is not asserted. Forced breakdown, Fefferman (C) and (D), is a different statement and is not adjudicated here.